

Problem:

Given a sorted array in ascending order, insert a new number into the array

while maintaining the sorted order.

We cannot use sort() after inserting; instead, we must find the correct position

and insert the element directly.

Example:

Input array: [1, 3, 5, 7, 9]

New number: 6

Expected output: [1, 3, 5, 6, 7, 9]

```
def insert_sorted(arr, num):
```

```
    # Find the correct position for insertion
```

```
    for i in range(len(arr)):
```

```
        if arr[i] > num:
```

```
            arr.insert(i, num) # Insert the number at the correct position
```

```
            return arr
```

```
    # If the number is greater than all elements, append it to the end
```

```
    arr.append(num)
```

```
    return arr
```

Testing the function with the first dataset

```
arr = [1, 3, 5, 7, 9]
```

```
num = 6
```

```
print(insert_sorted(arr, num)) # Output: [1, 3, 5, 6, 7, 9]
```

Applying the same logic to a different dataset:

Input array: [2, 4, 8, 10]

New number: 5

Expected output: [2, 4, 5, 8, 10]

```
arr2 = [2, 4, 8, 10]
```

```
num2 = 5
```

```
print(insert_sorted(arr2, num2)) # Output: [2, 4, 5, 8, 10]
```